Reconciling Treatment Effect Estimates across Levels of Aggregation

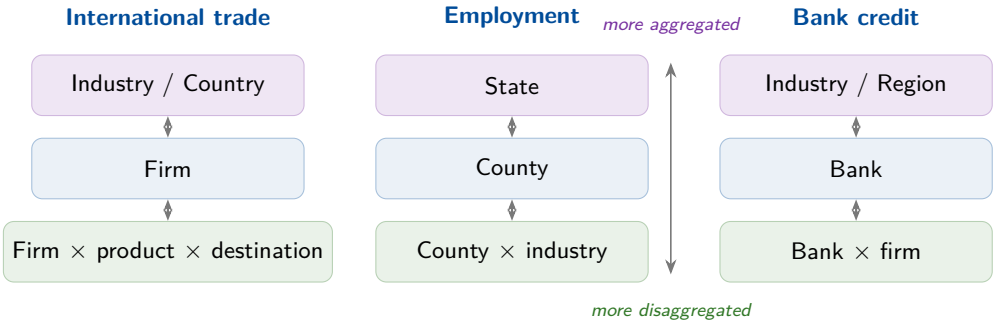
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System Committee on Econometrics 2026 Meeting
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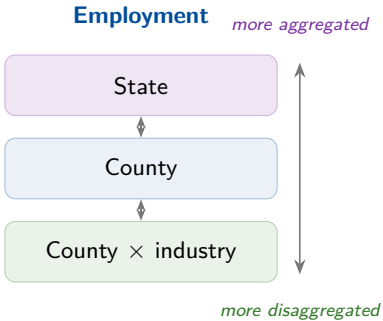
Economic data is naturally layered

Same underlying economic activity at different levels of observation



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This paper in a nutshell

1. Define disaggregation invariance

2. Define aggregation invariance

3. Run simulations to assess the magnitude of the biases
   → See paper

4. Show that the biases are sizable in a real setting: employment multiplier of government spending at the county × industry, county, and state levels
   → Replication publicly available at: github.com/paulhbeaumont/invariance-replication

Disaggregation/Aggregation used across many settings

- **Local labor market**: geography → geography × industry
  (Mian & Sufi 2014; Nakamura & Steinsson 2014; Guren et al. 2020; Kovak 2013)
- **Industry**: industry → industry × geography
  (Autor et al. 2013; Pierce & Schott 2016; Hombert & Matray 2018; Boehm et al. 2023)
- **Firm trade**: firm → firm × market
  (Matray et al. 2026; Mayer et al. 2014; Hummels et al. 2014; Paravisini et al. 2015; Bernard et al. 2011; Berman et al. 2012)
- **Bank credit**: bank → bank × firm
  (Khwaja & Mian 2008, Amiti & Weinstein 2011, 2018, Jiménez et al. 2012, 2014)
- **Multi-plant firms**: firm → firm × plant
  (Davis & Haltiwanger 1992, Abowd et al. 1999, Giroud & Mueller 2019, Cestone et al. 2024)
- **Product-market**: firm → industry / market
  (Garthwaite 2014; Helm 2020)
- **Local GE**: firm / county → region / labor market
  (Kline & Moretti 2014; Huber 2018; Giroud & Mueller 2019; Mian, Sarto & Sufi 2023)
- **Bank credit**: bank × firm → firm
  (Khwaja & Mian 2008; Cingano et al. 2016; Beck et al. 2021; Herreno 2023)
- **Fiscal multipliers**: county / region shocks → state or national response
  (Nakamura & Steinsson 2014, 2018; Guren et al. 2021)

Outline

Disaggregation Invariance
- Sources of non-invariance
- Solutions
- Example: Fiscal shocks and employment

Aggregation Invariance

Conclusion

Why do we disaggregate?

We often can measure the economic relationship we care about at lower levels

Firm exports	$\longrightarrow$	**Firm $\times$ destination $\times$ product**
County employment	$\longrightarrow$	**County $\times$ industry**
Bank credit	$\longrightarrow$	**Bank $\times$ firm**

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The economic objects we care about ("baseline")

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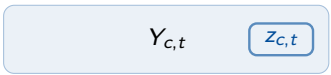
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Lower levels we do not care about per se, used only as an identification trick

$\longrightarrow$ Simply disaggregating the data can change the baseline elasticities even if the economics stay exactly the same

Causal impact of z on Y?

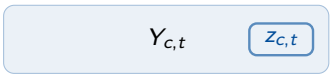
E.g.: Effect of government spending in a county (z_{ct}) on employment growth (Y_{ct})



Baseline: $\Delta y_{ct} = \beta^c z_{ct} + \varepsilon_{ct}$

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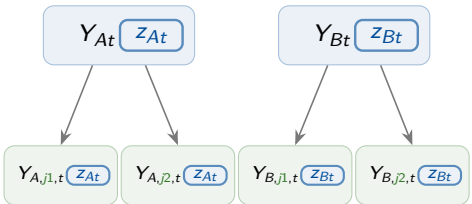


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Concern: industry shocks correlated with $z_{ct} \Rightarrow \beta^c$ biased

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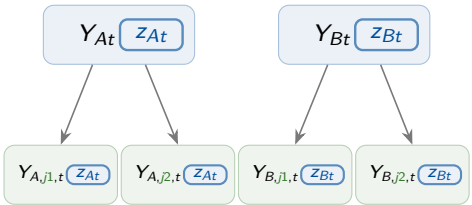
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Fix: disaggregated outcomes (same shock!) can absorb national industry trends: γ_{jt}

Disaggregated: $\Delta y_{cjt} = \beta^{c \times j} z_{ct} + \gamma_{jt} + \varepsilon_{ct}$

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Standard robustness check: is $\beta^{c \times j}$ still significant?

Disaggregate outcomes for identification

Goal: Absorb national industry shocks that may be correlated with the county shock z_{ct}

Baseline:

$$\Delta y_{ct} = \beta^{c} z_{ct} + \varepsilon_{ct}$$

Disaggregated outcomes:

$$\Delta y_{cjt} = \beta^{c \times j, \mathrm{FE}} z_{ct} + \gamma_{jt} + \varepsilon_{cjt}$$

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1. **Specification** changes: we can add γ_{jt}
2. **But outcome level** also changes: $y_c \to y_{cj}$

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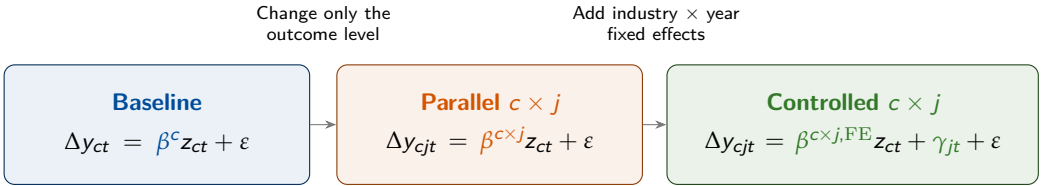
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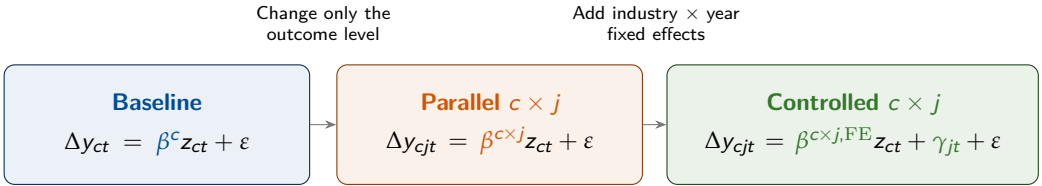
1. **Specification** changes: we can add γ_{jt}
2. **But outcome level** also changes: $y_c \rightarrow y_{cj}$

If the coefficient changes, is it because of the industry shock (#1) or because we changed the level of observation (#2)?

Disaggregation invariance: separate disaggregated outcomes from new controls

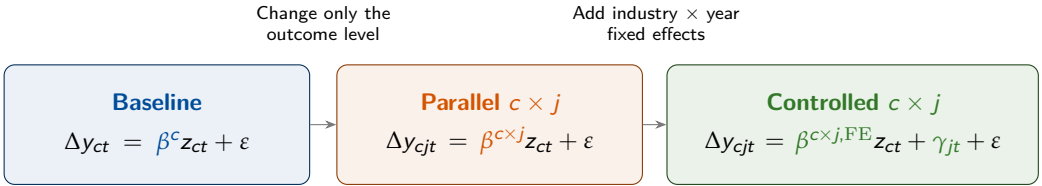


Disaggregation invariance: separate disaggregated outcomes from new controls



$$\underbrace{\beta^{c \times j, \mathrm{FE}} - \beta^{c}}_{\text{observed gap}} = \underbrace{\left(\beta^{c \times j} - \beta^{c}\right)}_{\substack{\text{mechanical} \\ \text{disaggregation}}} + \underbrace{\left(\beta^{c \times j, \mathrm{FE}} - \beta^{c \times j}\right)}_{\substack{\text{new-control} \\ \text{term}}}$$

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Disaggregation Invariance: $\beta^{c} = \beta^{c\times j}$

Mechanical term $= 0$ so the observed gap captures impact of new controls

Disaggregation invariance: separate disaggregated outcomes from new controls

Change only the outcome level

Add industry × year fixed effects

Baseline

$$\Delta y_{ct} = \beta^{c} z_{ct} + \varepsilon$$

Parallel $c \times j$

$$\Delta y_{cjt} = \beta^{c \times j} z_{ct} + \varepsilon$$

Controlled $c \times j$

$$\Delta y_{cjt} = \beta^{c \times j, \mathrm{FE}} z_{ct} + \gamma_{jt} + \varepsilon$$

$$\underbrace{\beta^{c \times j, \mathrm{FE}} - \beta^{c}}_{\text{observed gap}} = \underbrace{\left(\beta^{c \times j} - \beta^{c}\right)}_{\substack{\text{mechanical} \\ \text{disaggregation}}} + \underbrace{\left(\beta^{c \times j, \mathrm{FE}} - \beta^{c \times j}\right)}_{\substack{\text{new-control} \\ \text{term}}}$$

Usually does not hold

Changing only the outcome level frequently changes the estimate

Disaggregation invariance: $\beta^c = \beta^{c \times j}$

County-level regression: each county-year enters once

$$\Delta y_{ct} = \beta^c z_{ct} + \varepsilon_{ct}, \qquad \beta^c = \frac{\sum_{c,t} \tilde{z}_{ct} \Delta y_{ct}}{\sum_{c,t} \tilde{z}_{ct}^2}$$

$\tilde{z}_{ct}$: shock residualized on the controls, so each $\beta = \mathrm{Cov}/\mathrm{Var}$

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Parallel county $\times$ industry regression: the same county shock is copied across cells

$$\Delta y_{cjt} = \beta^{c \times j} z_{ct} + \varepsilon_{cjt},$$
$$\beta^{c \times j} = \frac{\sum_{c,t} \sum_{j} w_{cjt} \tilde{z}_{ct} \Delta y_{cjt}}{\sum_{c,t} \sum_{j} w_{cjt} \tilde{z}_{ct}^{2}}$$

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Invariance fails when the weighted covariances differ

Three reasons: **implicit weights**, **entry and exit**, and **nonlinear growth rates**

Estimand gap formally

Log growth on continuing county $\times$ industry cells, same shock z_{ct} in both regressions

$$\beta^{c} - \beta_{EW}^{c\times j} = \underbrace{\left(\beta_{VW}^{c\times j} - \beta_{EW}^{c\times j}\right)}_{\text{Implicit weights}} + \underbrace{\beta^{\mathcal{J}}}_{\text{Jensen}} + \underbrace{\left(\beta^{c} - \beta_{\text{cont}}^{c}\right)}_{\text{Entry and exit}}$$

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- $\beta_{EW}^{c\times j}$: equal-weighted cell regression, each cell counts once
- $\beta_{VW}^{c\times j}$: value-weighted cell regression, weighted by initial employment shares $s_{cjt} = Y_{cjt}/Y_{ct}^{\text{cont}}$, which sum to one within each county

Happen when treatment is heterogeneous:

- *Within county:* treatment raises employment more in a county's larger industries
- *Across counties:* counties with more industries respond more strongly to treatment

Estimand gap formally

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Very common in the data: disaggregated # cells, 75th/25th often > 10

[Implicit weights in the data]

Estimand gap formally

Log growth on continuing county × industry cells, same shock z_{ct} in both regressions

$$\beta^{c} - \beta_{EW}^{c\times j} = \underbrace{\left(\beta_{VW}^{c\times j} - \beta_{EW}^{c\times j}\right)}_{\text{Implicit weights}} + \underbrace{\beta^{\mathcal{J}}}_{\text{Jensen}} + \underbrace{\left(\beta^{c} - \beta_{\text{cont}}^{c}\right)}_{\text{Entry and exit}}$$

Jensen term of county c:

$$\mathcal{J}_{ct} = \Delta y_{ct}^{\text{cont}} - \sum_{j} s_{cjt}\,\Delta y_{cjt} \approx \tfrac{1}{2}\,\text{Var}_{s}(\Delta y_{cjt})$$

- Binary treatment: $\beta^{\mathcal{J}} = E[\mathcal{J}_{ct} \mid T = 1] - E[\mathcal{J}_{ct} \mid T = 0]$, the difference in the Jensen term between treated and control counties
- Continuous shock z_{ct} (our application): same object as a slope, $\beta^{\mathcal{J}} = \text{Cov}(\mathcal{J}_{ct}, \tilde{z}_{ct})/\text{Var}(\tilde{z}_{ct})$

Estimand gap formally

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β_{cont}^{c}: county regression on $\Delta y_{ct}^{\text{cont}}$, log growth of employment in continuing industries

Gap increases if the treatment affects entry and/or exit in different proportions

Very common in the data: extensive margin often $>$ 50% of activity at $h = 3$ [Extensive margin in the data]

[Measuring extensive margins in growth formula]

Outline

Disaggregation Invariance
- Sources of non-invariance
- Solutions
- Example: Fiscal shocks and employment

Aggregation Invariance

Conclusion

Solutions

1. Midpoint Growth Rate + Midpoint Weights [Numerical and Empirical Illustrations]

$$y_{ij,t}^{mp} = 2 \cdot \frac{y_{ij,t} - y_{ij,t-1}}{y_{ij,t} + y_{ij,t-1}} \qquad \omega_{ij,t}^{mp} = \frac{y_{ij,t} + y_{ij,t-1}}{y_{i,t} + y_{i,t-1}}$$

- Defined for zeros: entry ($\Delta = +2$), exit ($\Delta = -2$)
  $\rightarrow$ *keeps extensive margin*
- Aggregates linearly: $y_{i,t}^{mp} = \sum_j \omega_{ij,t}^{mp} y_{ij,t}^{mp}$
  e.g., county MPGR = weighted average of county-industry MPGRs
  $\rightarrow$ *No Jensen's inequality*
- Sub-units weighted by contribution to baseline unit
  $\rightarrow$ *corrects implicit weighting*

2. Poisson [Details]

Pros and cons of Poisson vs MPGR

	MPGR	**Poisson**
Zeros	✓ for disaggregation (entry/exit across sub-units) ✗ **not** at the level of the shock → overweights extensive margin (ongoing work)	✓ natively But can produce puzzling results (e.g., both margins +, combined = 0)
Fixed effects	Spec in growth → unit FE differenced out Scales well to very large datasets	Spec in levels → requires all unit FE Computationally heavy for large datasets
Margin decomp.	Exact	Approximation

Bottom line: both give disaggregation invariance; prefer MPGR for large panels (growth spec, unit FE difference out) and the exact margin split; prefer Poisson when zeros *at the level of the shock* matter

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Empirical setting: Fiscal shocks and employment

Question: Impact of military spending on county employment? (Nakamura and Steinsson 2014)

Data: 3,058 county groups, 1981–2016

- Military procurement: DCADS (National Archives) and USAspending; real net contract obligations
- Employment: County Business Patterns, Eckert, Fort, Schott and Yang (2021); county $\times$ industry cells
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Treatment: change in average annual procurement per resident over three years

$$x_{ct} = \frac{\overline{S}_{ct} - \overline{S}_{c,t-3}}{\mathrm{Pop}_{c,t-3}}$$

$\overline{S}_{ct}$: average annual procurement over $t-2$ to t

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Instrument: historical procurement intensity × national change in spending

$$z_{ct} = \underbrace{a_{ct}}_{\substack{\text{county procurement per resident} \\ \text{in } t-12 \text{ to } t-8}} \times \underbrace{n_t}_{\substack{\text{national change in} \\ \text{spending per resident}}}$$

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A national military buildup raises procurement more in counties that historically received more of it

Specifications to test disaggregation invariance

County level

$$\Delta y_{ct} = \beta^{c} x_{ct} + \alpha_c + \delta_t + \varepsilon_{ct}$$

- Outcome: employment growth for county c
- x_{ct}: county spending, instrumented by z_{ct} (2SLS)
- α_c, δ_t: county and year FE

County $\times$ industry level

$$\Delta y_{cjt} = \beta^{c \times j} x_{ct} + \alpha_c + \delta_t + \varepsilon_{cjt}$$

- Outcome: employment growth in county c, industry j
- Same county spending copied to each county-industry cell

Disaggregation invariance: $\beta^{c} = \beta^{c \times j}$

Results

– Log growth at lagged employment shares misses the county by −16.7% at NAICS2 and −34.9% at NAICS6

	Log growth of employment		
Weights	Lagged employment share		
Level	County	County × NAICS2	County × NAICS6
	(1)	(2)	(3)
County spending	6.123**	5.104*	3.984
	(2.548)	(2.610)	(2.695)
% deviation from county	0.0%	-16.7%	-34.9%
Fixed effects			
County, year	✓	✓	✓
Industry × year			
First-stage F	33.41	33.30	32.46
Observations	110,088	2,359,648	22,506,999

2SLS with the exposure instrument z_{ct}; three-year growth; coefficients $\times 10^5$; county-clustered standard errors

Results

– With WLS midpoint growth, same economics = same coefficients

	Log growth of employment			**Midpoint growth** of employment		
Weights	Lagged employment share			Midpoint share		
Level	County	County × NAICS2	County × NAICS6	County	County × NAICS2	County × NAICS6
	(1)	(2)	(3)	(5)	(6)	(7)
County spending	6.123**	5.104*	3.984	6.353***	6.353***	6.353***
	(2.548)	(2.610)	(2.695)	(2.397)	(2.397)	(2.397)
% deviation from county	0.0%	-16.7%	-34.9%	0.0%	0.0%	0.0%
Fixed effects						
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Industry × year						
First-stage F	33.41	33.30	32.46	33.41	33.41	33.41
Observations	110,088	2,359,648	22,506,999	110,088	2,551,814	36,517,233

2SLS with the exposure instrument z_{ct}; three-year growth; coefficients $\times 10^5$; county-clustered standard errors

From county to county × industry log-growth coefficients

Log growth at lagged shares: County (col. 1) to County × NAICS6 (col. 3), before industry × year FE



Green bars: increases; red bars: decreases; coefficients $\times 10^5$

Results

– Including Industry × year leads to qualitatively opposite conclusions

	Log growth of employment				**Midpoint growth** of employment		
Weights	Lagged employment share				Midpoint share		
Level	County	County × NAICS2	County × NAICS6	County × NAICS6	County	County × NAICS2	County × NAICS6
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
County spending	**6.123****	5.104*	3.984	**2.731**	6.353***	6.353***	6.353***
	(2.548)	(2.610)	(2.695)	(1.902)	(2.397)	(2.397)	(2.397)
% deviation from county	0.0%	-16.7%	-34.9%	-55.4%	0.0%	0.0%	0.0%
Fixed effects							
County, year	✓	✓	✓	✓	✓	✓	✓
Industry × year				✓			
First-stage F	33.41	33.30	32.46	41.85	33.41	33.41	33.41
Observations	110,088	2,359,648	22,506,999	22,506,999	110,088	2,551,814	36,517,233

2SLS with the exposure instrument z_{ct}; three-year growth; coefficients $\times 10^5$; county-clustered standard errors

Results

– Including Industry × year leads to qualitatively opposite conclusions

	Log growth of employment				**Midpoint growth** of employment			
Weights	Lagged employment share				Midpoint share			
Level	County	County × NAICS2	County × NAICS6	County × NAICS6	County	County × NAICS2	County × NAICS6	County × NAICS6
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
County spending	6.123**	5.104*	3.984	2.731	**6.353*****	6.353***	6.353***	**5.388****
	(2.548)	(2.610)	(2.695)	(1.902)	(2.397)	(2.397)	(2.397)	(2.722)
% deviation from county	0.0%	-16.7%	-34.9%	-55.4%	0.0%	0.0%	0.0%	-15.2%
Fixed effects								
County, year	✓	✓	✓	✓	✓	✓	✓	✓
Industry × year				✓				✓
First-stage F	33.41	33.30	32.46	41.85	33.41	33.41	33.41	27.41
Observations	110,088	2,359,648	22,506,999	22,506,999	110,088	2,551,814	36,517,233	36,517,233

2SLS with the exposure instrument z_{ct}; three-year growth; coefficients $\times 10^5$; county-clustered standard errors

More in the paper

- **Simulations quantify magnitude of each deviation** [Details]
  - A two-level model lets treatment move both the number of active sub-units and activity per sub-unit
  - We switch on one channel at a time: entry/exit, Jensen's inequality, and implicit weights
  - Across the simulations, log growth can move mechanically with the level; MPGR with the right weights recovers the invariant benchmark

- **Practical guidance**

Outline

Disaggregation Invariance

Aggregation Invariance
- Estimating spillovers
- Example: Fiscal shocks and employment spillovers

Conclusion

Aggregate outcomes and treatment to detect spillovers

Identifying spillovers in the data requires specifying the higher level group at which they happen

Mechanism	Across units	Within a group
Business stealing	Exporters	Destination market
Positive externalities	Firms	Industry
Labor mobility	Counties	State

Intuitively, comparing levels comes from an accounting identity

$$\underbrace{Y_{st}}_{\text{State employment}} = \underbrace{\sum_{c \in \mathcal{T}_{st}} Y_{ct}}_{\text{Treated counties}} + \underbrace{\sum_{c \in s \setminus \mathcal{T}_{st}} Y_{ct}}_{\text{Untreated counties}}$$

Treatment in one county can affect employment elsewhere in the state

Usual comparison does not isolate the spillover

Raw county regression

$$\Delta y_{ct} = \beta^{c,\text{raw}} z_{ct} + \varepsilon_{ct}$$

Raw state regression

$$\Delta y_{st} = \beta^{s,\text{raw}} z_{st} + \varepsilon_{st}$$

Usual interpretation

$$\beta^{s,\text{raw}} - \beta^{c,\text{raw}} \quad \longrightarrow \quad \text{spillovers across counties within states}$$

Usual comparison does not isolate the spillover

Raw county regression

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Usual interpretation

$$\beta^{s,\text{raw}} - \beta^{c,\text{raw}} \longrightarrow \text{spillovers across counties within states}$$

False under most standard specifications

How do we align county and state growth?

Raw county regression

$$\Delta y_{ct} = \beta^{c,\text{raw}} z_{ct} + \varepsilon_{ct}$$

Raw state regression

$$\Delta y_{st} = \beta^{s,\text{raw}} z_{st} + \varepsilon_{st}$$

Required: state growth is linear in county growth

$$\Delta y_{st} = \sum_{c \in s} \omega_{ct}^{Y} \Delta y_{ct}$$

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$$\Delta y_{st} = \sum_{c \in s} \omega_{ct}^{Y} \Delta y_{ct}$$

Standard growth aggregates exactly

$$\frac{Y_{s,t+h} - Y_{st}}{Y_{st}} = \sum_{c \in s} \frac{Y_{ct}}{Y_{st}} \frac{Y_{c,t+h} - Y_{ct}}{Y_{ct}}$$

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$$\frac{Y_{s,t+h} - Y_{st}}{Y_{st}} = \sum_{c \in s} \frac{Y_{ct}}{Y_{st}} \frac{Y_{c,t+h} - Y_{ct}}{Y_{ct}}$$

Log growth leaves an aggregation gap

$$\log \frac{Y_{s,t+h}}{Y_{st}} \neq \sum_{c \in s} \frac{Y_{ct}}{Y_{st}} \log \frac{Y_{c,t+h}}{Y_{ct}}$$

The growth rate on the left-hand side must be linear in county growth

Why not use midpoint growth?

Midpoint growth is also linear in county growth

$$g_{st}^{MP} = \sum_{c \in s} \omega_{ct}^{MP} g_{ct}^{MP}, \qquad \omega_{ct}^{MP} = \frac{Y_{ct} + Y_{c,t+h}}{Y_{st} + Y_{s,t+h}}$$

Why not use midpoint growth?

Midpoint growth is also linear in county growth

$$g_{st}^{MP} = \sum_{c \in s} \omega_{ct}^{MP} g_{ct}^{MP}, \qquad \omega_{ct}^{MP} = \frac{Y_{ct} + Y_{c,t+h}}{Y_{st} + Y_{s,t+h}}$$

Its weights contain post-treatment outcomes

That matters once the same weights build the state shock in Step 2

How should we weight the county regression?

Raw county regression

$$\Delta y_{ct} = \beta^{c,\text{raw}} z_{ct} + \varepsilon_{ct}$$

Raw state regression

$$\Delta y_{st} = \beta^{s,\text{raw}} z_{st} + \varepsilon_{st}$$

How much does each county contribute to the state outcome?

How should we weight the county regression?

Raw county regression

$$\Delta y_{ct} = \beta^{c,\text{raw}} z_{ct} + \varepsilon_{ct}$$

Raw state regression

$$\Delta y_{st} = \beta^{s,\text{raw}} z_{st} + \varepsilon_{st}$$

Each county's growth enters in proportion to its initial employment share

$$\underbrace{\Delta y_{st}}_{\text{State growth}} = \sum_{c \in s} \omega_{ct}^{Y} \underbrace{\Delta y_{ct}}_{\text{County growth}}$$

For standard employment growth

$$\omega_{ct}^{Y} = \frac{Y_{ct}}{Y_{st}}$$

County's share of state employment before the shock

How should we weight the county regression?

Weighted county regression

$$\Delta y_{ct} = \beta^{c,W} z_{ct} + \varepsilon_{ct}$$

To match those contributions, weight the county regression by initial employment shares

W: weighted by initial employment shares

$$\beta^{c,\text{raw}} \longrightarrow \beta^{c,W}$$

Raw state regression

$$\Delta y_{st} = \beta^{s,\text{raw}} z_{st} + \varepsilon_{st}$$

Each county's growth enters in proportion to its initial employment share

$$\underbrace{\Delta y_{st}}_{\text{State growth}} = \sum_{c \in s} \omega_{ct}^{Y} \underbrace{\Delta y_{ct}}_{\text{County growth}}$$

For standard employment growth

$$\omega_{ct}^{Y} = \frac{Y_{ct}}{Y_{st}}$$

County's share of state employment before the shock

How do you construct the state shock?

Weighted county regression

$$\Delta y_{ct} = \beta^{c,W} z_{ct} + \varepsilon_{ct}$$

Raw state regression

$$\Delta y_{st} = \beta^{s,\text{raw}} z_{st} + \varepsilon_{st}$$

The outcome determines the treatment-aggregation weights

$$z_{st}^{Y} = \sum_{c \in s} \omega_{ct}^{Y} z_{ct}, \qquad \omega_{ct}^{Y} = \frac{Y_{ct}}{Y_{st}}$$

How do you construct the state shock?

Weighted county regression

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$$\Delta y_{st} = \beta^{s,\text{raw}} z_{st} + \varepsilon_{st}$$

The outcome determines the treatment-aggregation weights

$$z_{st}^{Y} = \sum_{c \in s} \omega_{ct}^{Y} z_{ct}, \qquad \omega_{ct}^{Y} = \frac{Y_{ct}}{Y_{st}}$$

Common practices that break alignment

- **County** $\to$ **State,** $Y =$ **Employment**:
  population shares rather than initial employment shares
- **Firm** $\to$ **Industry,** $Y =$ **Exports**:
  employment or total-sales shares rather than initial export shares
- **Bank** $\to$ **City,** $Y =$ **Credit**:
  deposit shares rather than initial loan shares

How do you construct the state shock?

Weighted county regression

$$\Delta y_{ct} = \beta^{c,W} z_{ct} + \varepsilon_{ct}$$

Raw state regression

$$\Delta y_{st} = \beta^{s,\text{raw}} z_{st} + \varepsilon_{st}$$

The outcome determines the treatment-aggregation weights

$$z_{st}^{Y} = \sum_{c \in s} \omega_{ct}^{Y} z_{ct}, \qquad \omega_{ct}^{Y} = \frac{Y_{ct}}{Y_{st}}$$

Another form of introducing misalignment

Summing the underlying variables and reconstructing the state shock can impose weights that differ from the outcome's weights

$$\underbrace{z_{ct} = \frac{S_{ct}}{P_{ct}}}_{\text{Spending per capita}}, \qquad \frac{\sum_{c \in s} S_{ct}}{\sum_{c \in s} P_{ct}} = \sum_{c \in s} \underbrace{\frac{P_{ct}}{P_{st}}}_{\text{Population shares}} z_{ct}$$

Summing spending and population first weights counties by their population shares

Why not use midpoint weights?

With midpoint growth on the left-hand side, the outcome's weights are ω_{ct}^{MP}

But the corresponding exposure uses post-treatment outcomes

$$z_{st}^{MP} = \sum_{c \in s} \omega_{ct}^{MP} z_{ct}, \qquad \omega_{ct}^{MP} = \frac{Y_{ct} + Y_{c,t+h}}{Y_{st} + Y_{s,t+h}}$$

Treatment can change these weights, making state exposure outcome-dependent

Within and between variations

Weighted county regression

$$\Delta y_{ct} = \beta^{c,W} z_{ct} + \varepsilon_{ct}$$

Aligned state regression

$$\Delta y_{st} = \beta^{s} z^{Y}_{st} + \varepsilon_{st}$$

$$\beta^{s} - \beta^{c,W} \stackrel{?}{=} \text{spillovers across counties within states}$$

We have aligned the weights, state exposure, and growth rates

Within and between variations

Weighted county regression

$$\Delta y_{ct} = \beta^{c,W} z_{ct} + \varepsilon_{ct}$$

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$$\Delta y_{st} = \beta^{s} z^{Y}_{st} + \varepsilon_{st}$$

$$\beta^{s} - \beta^{c,W} \stackrel{?}{=} \text{spillovers across counties within states}$$

We have aligned the weights, state exposure, and growth rates

Weighted county regression: across counties within and across states $\longrightarrow$ We still need to separate own-county treatment from state exposure

Within and between variations

Weighted county regression

$$\Delta y_{ct} = \beta^{c,W} z_{ct} + \varepsilon_{ct}$$

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We have aligned the weights, state exposure, and growth rates

Weighted county regression: across counties within and across states $\longrightarrow$ We still need to separate own-county treatment from state exposure

Intended comparison: across counties within states

Align the county regression with the state comparison

Option 1: Add state exposure

$$\Delta y_{ct} = \underbrace{\beta^{c,W}_{\text{aug}} z_{ct}}_{\substack{\text{Own-county response} \\ \text{holding state exposure fixed}}} + \underbrace{\delta z^{Y}_{st}}_{\substack{\text{State-exposure response} \\ \text{holding own treatment fixed}}} + \varepsilon_{ct}$$

County regression: WLS with initial employment share $\omega^{Y}_{ct} = Y_{ct}/Y_{st}$

– **Control for state exposure** to separate it from own-county treatment

Align the county regression with the state comparison

Option 2: Add state × year FE

$$\Delta y_{ct} = \underbrace{\beta^{c,W}_{\text{aug,FE}} z_{ct}}_{\substack{\text{Own-county response} \\ \text{within state-year}}} + \underbrace{\gamma_{st}}_{\substack{\text{State×year FE} \\ \text{absorb state exposure}}} + \varepsilon_{ct}$$

County regression: WLS with initial employment share $\omega^{Y}_{ct} = Y_{ct}/Y_{st}$

- Compare state-level β^{s} with county-level $\beta^{c,W}_{\text{aug,FE}}$
- **Equivalence theorem:** $\beta^{s} - \beta^{c,W}_{\text{aug,FE}} = \delta$

When can we interpret δ as a spillover?

$$\Delta y_{ct} = \beta^{c,W}_{\text{aug}} z_{ct} + \underbrace{\delta}_{\text{State-exposure response}} z^{Y}_{st} + \varepsilon_{ct}$$

State exposure requires its own exogeneity condition

Exogenous county treatment does not imply exogenous state exposure

Required: state exposure is unrelated to other determinants of county growth, conditional on own treatment and controls

When can we interpret δ as a spillover?

$$\Delta y_{ct} = \beta^{c,W}_{\text{aug}} z_{ct} + \underbrace{\delta}_{\text{State-exposure response}} z^{Y}_{st} + \varepsilon_{ct}$$

State exposure requires its own exogeneity condition

Exogenous county treatment does not imply exogenous state exposure

Required: state exposure is unrelated to other determinants of county growth, conditional on own treatment and controls

Different treatment effects can generate composition effects

Within-state and across-state comparisons can load differently on counties with different treatment effects

Sufficient condition: homogeneous own-treatment effects rule out this composition channel

Aligning the regressions makes the comparison meaningful
These additional conditions give δ a spillover interpretation

Outline

Disaggregation Invariance

Aggregation Invariance
- Estimating spillovers
- Example: Fiscal shocks and employment spillovers

Conclusion

Results: Spillovers from fiscal spending shocks

Direct estimate: spillover in the augmented county regression (weighted 2SLS)

$$\Delta y_{ct} = \beta^{c,W}_{\text{aug}} z_{ct} + \delta\, z^{Y}_{st} + \varepsilon_{ct}$$

	Direct	
	δ	First-stage F
State (51)	8.076**	41.86
	(3.263)	
EA (179)	3.953*	4.59
	(2.301)	
CEA (343)	7.763*	3.43
	(4.334)	

SEs clustered by aggregate unit; standard three-year growth; coefficients $\times 10^5$ (pp per \$1,000 per resident). First-stage F for the aggregate exposure in the augmented county regression.

Commuting zones (736) not shown: $\hat{\delta} = -13.002$ (12.180)

Results: Spillovers from fiscal spending shocks

Aligned coefficient-gap check: state regression $\Delta y_{st} = \beta^s z^Y_{st} + \varepsilon_{st}$

$$\widehat{\beta}^s - \widehat{\beta}^{c,W}_{\text{aug}} = 10.830 - 2.664 = 8.166 \approx \widehat{\delta} = 8.076$$

Residual 0.091 (0.84% of $\widehat{\beta}^s$) because employment shares vary over time

	Direct		Aligned gap check	
	δ	First-stage F	β^s	$\beta^{c,W}_{\text{aug}}$
State (51)	8.076**	41.86	10.830***	2.664***
	(3.263)		(3.063)	(0.981)
EA (179)	3.953*	4.59	7.031***	3.026***
	(2.301)		(2.406)	(0.958)
CEA (343)	7.763*	3.43	10.964**	3.160***
	(4.334)		(4.339)	(1.043)

SEs clustered by aggregate unit; standard three-year growth; coefficients $\times 10^5$ (pp per \$1,000 per resident). First-stage F for the aggregate exposure in the augmented county regression.

Commuting zones (736) not shown: $\widehat{\delta} = -13.002$ (12.180)

Results: Spillovers from fiscal spending shocks

Naive comparison: drop state exposure from the county regression

$$\widehat{\delta}^{\text{naive}} = \widehat{\beta}^{s} - \widehat{\beta}^{c,W}$$

	Direct		Aligned gap check		Naive
	δ	First-stage F	β^{s}	$\beta^{c,W}_{\text{aug}}$	δ^{naive}
State (51)	8.076**	41.86	10.830***	2.664***	5.446**
	(3.263)		(3.063)	(0.981)	(2.566)
EA (179)	3.953*	4.59	7.031***	3.026***	2.376*
	(2.301)		(2.406)	(0.958)	(1.337)
CEA (343)	7.763*	3.43	10.964**	3.160***	3.714
	(4.334)		(4.339)	(1.043)	(2.310)

SEs clustered by aggregate unit; standard three-year growth; coefficients $\times 10^5$ (pp per \$1,000 per resident). First-stage F for the aggregate exposure in the augmented county regression.

Commuting zones (736) not shown: $\widehat{\delta} = -13.002$ (12.180)

Results: Spillovers from fiscal spending shocks

Naive comparison: drop state exposure from the county regression

$$\widehat{\delta}^{\text{naive}} = \widehat{\beta}^{s} - \widehat{\beta}^{c,W}$$

The naive comparison understates the spillover by 33% to 52%

	Direct		Aligned gap check		Naive	Understated by
	δ	First-stage F	β^{s}	$\beta^{c,W}_{\text{aug}}$	δ^{naive}	$1 - \delta^{\text{naive}}/\delta$
State (51)	8.076**	41.86	10.830***	2.664***	5.446**	33%
	(3.263)		(3.063)	(0.981)	(2.566)	
EA (179)	3.953*	4.59	7.031***	3.026***	2.376*	40%
	(2.301)		(2.406)	(0.958)	(1.337)	
CEA (343)	7.763*	3.43	10.964**	3.160***	3.714	52%
	(4.334)		(4.339)	(1.043)	(2.310)	

SEs clustered by aggregate unit; standard three-year growth; coefficients $\times 10^5$ (pp per \$1,000 per resident). First-stage F for the aggregate exposure in the augmented county regression.

Commuting zones (736) not shown: $\widehat{\delta} = -13.002$ (12.180)

Results: Role of each source of deviation

Alignment: large
- Naive comparison: measured spillover 33% to 52% too small
  - State: 5.446 vs 8.076; EA: 2.376 vs 3.953; CEA: 3.714 vs 7.763

Data: mixed
- Winsorization: negligible here
  - Other papers: deviations **1-2x larger** than baseline
- Naive shock aggregation: **large**, raising β^s by 34%

Estimator: small relative to the state baseline $\widehat{\delta} = 8.076$
- Implicit weights: drop county weights, $\widehat{\delta} = 7.559$ (-6%)
- Jensen's inequality: log growth, $\widehat{\delta} = 7.680$ (-5%)

Extensive margin: none
- Counties with positive employment throughout: no entry or exit by construction

Horizon: similar spillover at $h = 2$ (6.271) and $h = 4$ (8.870)

Outline

Disaggregation Invariance

Aggregation Invariance

Conclusion

Most empirical research uses different levels of (dis)aggregated data

Existing work:

- Does not preserve disaggregation invariance
- Misattributes aggregation non-invariance

This paper: Easily implementable fixes makes it possible to interpret regressions

Most empirical research uses different levels of (dis)aggregated data

Existing work:

- Does not preserve disaggregation invariance
- Misattributes aggregation non-invariance

This paper: Easily implementable fixes makes it possible to interpret regressions

Disaggregation: Use midpoint growth rate + weights or Poisson
- Easy to check that this has been done correctly

Aggregation: Use augmented regression + standard growth rates + WLS + correct cleaning
- Differences between levels can be interpreted as spillovers

- We are happy to help to apply this with you on a live project!

Thank you!

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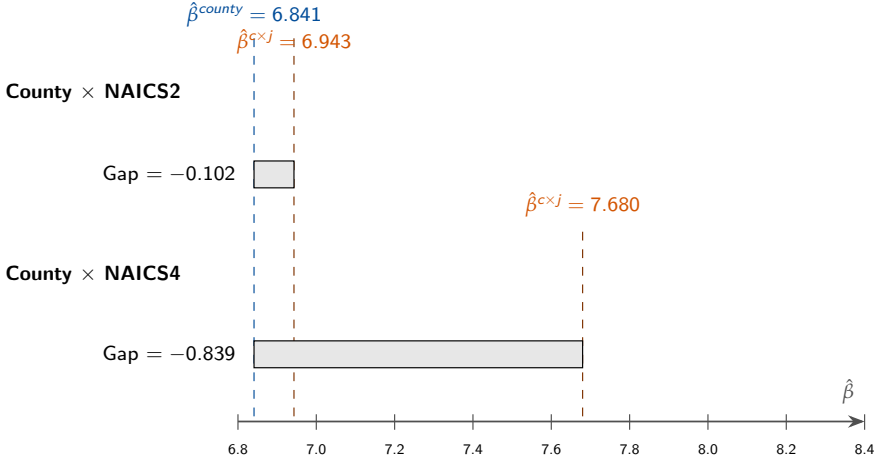
Appendix

Results: standard fiscal shock

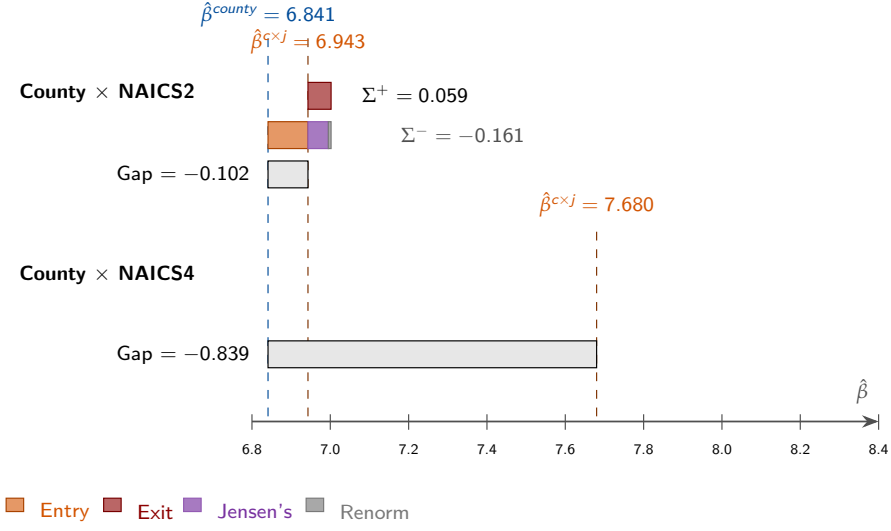
	Log growth of employment			**Midpoint growth** of employment		
Weights	Lagged employment share			Midpoint share		
Level	County	County × NAICS2	County × NAICS4	County	County × NAICS2	County × NAICS4
	(1)	(2)	(3)	(5)	(6)	(7)
Fiscal spending	6.841***	6.943***	7.680***	6.837***	6.837***	6.837***
	(1.469)	(1.417)	(1.202)	(1.466)	(1.466)	(1.466)
gap from county		+0.102	+0.839			
Fixed effects						
County + year	✓	✓	✓	✓	✓	✓
Industry × year						
Observations	107,030	2,284,807	12,520,809	107,030	2,782,780	32,430,090

– Log at lagged shares differs from the county by +0.102 at NAICS2 and +0.839 at NAICS4

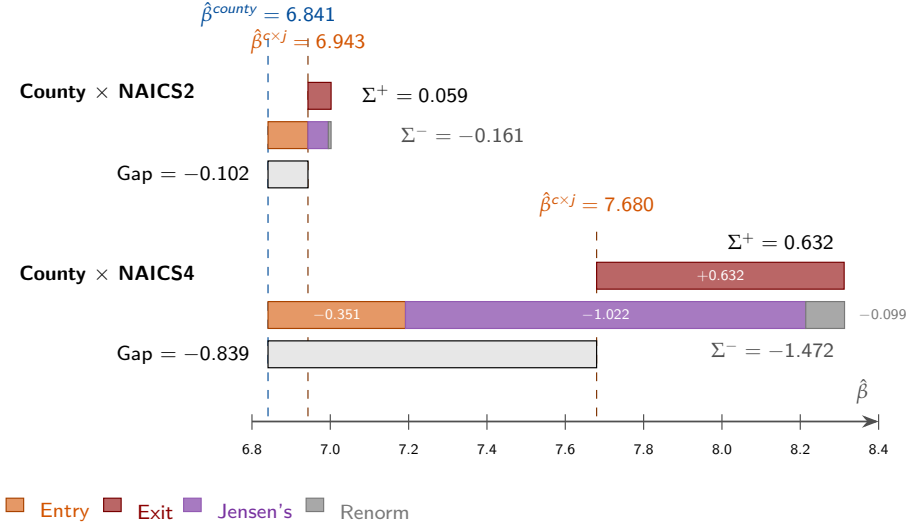
Decomposing the log growth gap: standard fiscal shock



Decomposing the log growth gap: standard fiscal shock



Decomposing the log growth gap: standard fiscal shock



Results: standard fiscal shock

	Log growth of employment				**Midpoint growth** of employment		
Weights	Lagged employment share				Midpoint share		
Level	County	County × NAICS2	County × NAICS4	County × NAICS4	County	County × NAICS2	County × NAICS4
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Fiscal spending	**6.841*****	6.943***	7.680***	**7.344*****	6.837***	6.837***	6.837***
	(1.469)	(1.417)	(1.202)	(1.380)	(1.466)	(1.466)	(1.466)
gap from county		+0.102	+0.839	+0.503			
Fixed effects							
County + year	✓	✓	✓	✓	✓	✓	✓
Industry × year				✓			
Observations	107,030	2,284,807	12,520,809	12,520,809	107,030	2,782,780	32,430,090

- Log at lagged shares differs from the county by +0.102 at NAICS2 and +0.839 at NAICS4
- Including Industry × year leaves a +0.503 log gap and a +0.130 MPGR gap

Results: standard fiscal shock

	Log growth of employment				**Midpoint growth** of employment			
Weights	Lagged employment share				Midpoint share			
Level	County	County × NAICS2	County × NAICS4	County × NAICS4	County	County × NAICS2	County × NAICS4	County × NAICS4
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Fiscal spending	6.841***	6.943***	7.680***	7.344***	**6.837***	6.837***	6.837***	**6.967***
	(1.469)	(1.417)	(1.202)	(1.380)	(1.466)	(1.466)	(1.466)	(1.729)
gap from county		+0.102	+0.839	+0.503				+0.130
Fixed effects								
County + year	✓	✓	✓	✓	✓	✓	✓	✓
Industry × year				✓				✓
Observations	107,030	2,284,807	12,520,809	12,520,809	107,030	2,782,780	32,430,090	17,387,589

- Log at lagged shares differs from the county by +0.102 at NAICS2 and +0.839 at NAICS4
- Including Industry × year leaves a +0.503 log gap and a +0.130 MPGR gap

Channel 1 numerical example: equal weighting flips the sign

County A has 1 industry (+40%); County B has 3 industries (each −20%); Control County C has 0 growth

Panel A. County level

County	$g^{emp}_{c,t+h}$
A	+0.40
B	−0.20
C	0.00
$\hat{\beta}^{\text{county}}$	+0.10 ✓

EW-OLS: $\dfrac{+0.40 - 0.20}{2} = +0.10$

Panel B. County × industry

County	Industry	$g^{emp}_{c,j,t+h}$
A	j_1	+0.40
B	j_1	−0.20
B	j_2	−0.20
B	j_3	−0.20
C	j_1	0.00
$\hat{\beta}^{\text{c-j}}$		−0.05 ✗

EW-OLS: $\dfrac{+0.40 - 3(0.20)}{4} = -0.05$

Channel 1 numerical example: equal weighting flips the sign

County A has 1 industry (+40%); County B has 3 industries (each −20%); Control County C has 0 growth

Panel A. County level

County	$g^{emp}_{c,t+h}$
A	+0.40
B	−0.20
C	0.00
$\hat{\beta}^{\text{county}}$	+0.10 ✓

EW-OLS: $\dfrac{+0.40 - 0.20}{2} = +0.10$

Panel B. County × industry

County	Industry	$g^{emp}_{c,j,t+h}$	
A	j_1	+0.40	
B	j_1	−0.20	3× weight
B	j_2	−0.20	
B	j_3	−0.20	
C	j_1	0.00	
$\hat{\beta}^{\text{c-j}}$		−0.05 ✗	

WLS-OLS: $\dfrac{+0.40 - 3 \cdot \frac{1}{3}(0.20)}{2} = +0.10$ ✓

Channel 1 numerical example: equal weighting flips the sign

County A has 1 industry (+40%); County B has 3 industries (each −20%); Control County C has 0 growth

Panel A. County level

County	$g^{emp}_{c,t+h}$
A	+0.40
B	−0.20
C	0.00
$\hat{\beta}^{\text{county}}$	+0.10 ✓

EW-OLS: $\dfrac{+0.40 - 0.20}{2} = +0.10$

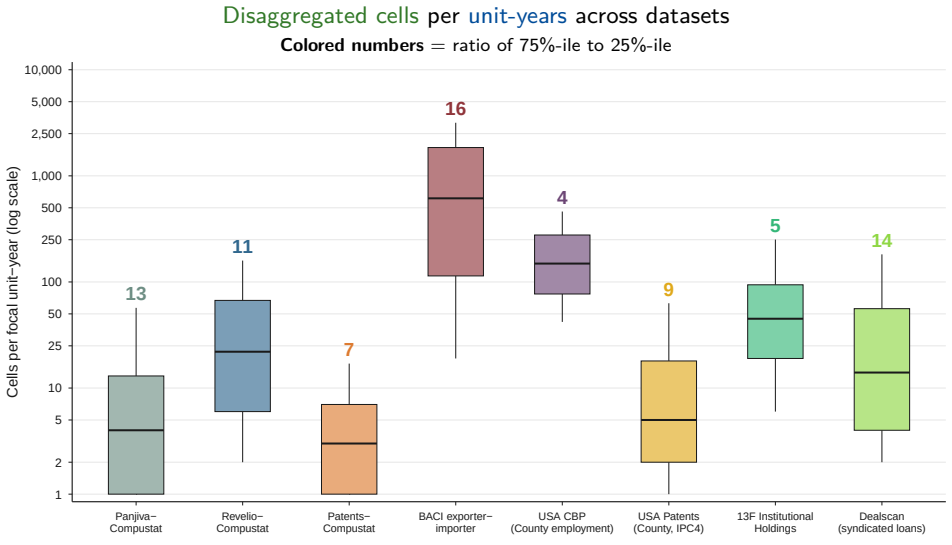
Panel B. County × industry

County	Industry	$g^{emp}_{c,j,t+h}$	
A	j_1	+0.40	
B	j_1	−0.20	3×1/3 weight
B	j_2	−0.20	
B	j_3	−0.20	
C	j_1	0.00	
$\hat{\beta}^{\text{c-j}}$		−0.05 ✗	

WLS-OLS: $\dfrac{+0.40 - 3 \cdot \frac{1}{3}(0.20)}{2} = +0.10$ ✓

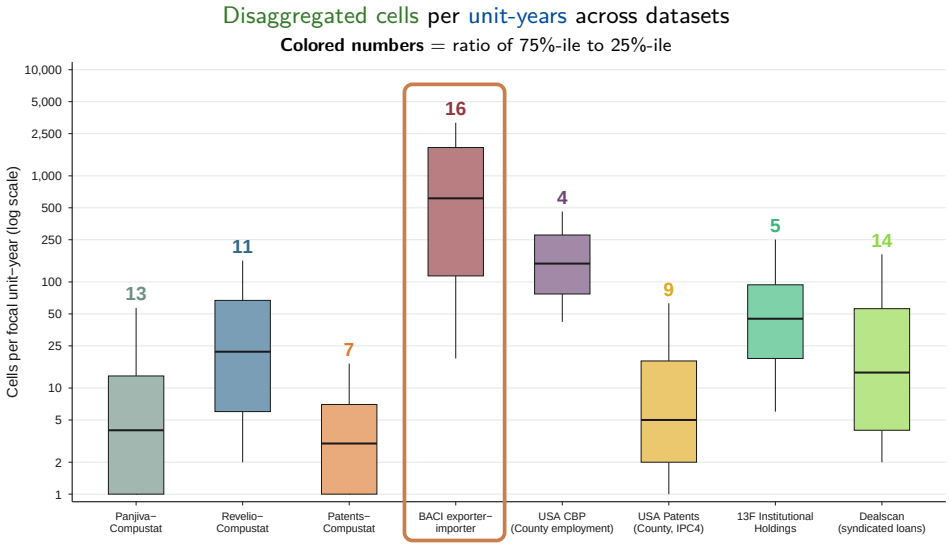
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Channel 1 in the data: Some units contribute many more cells



[Back]

Channel 1 in the data: Some units contribute many more cells



[Back]

Channel 2: Intensive vs Extensive margin effects

Write county employment growth as

$$\Delta y_{ct} = \sum_{j \in \mathcal{I}_c} I_{c,j} + \sum_{j \in \mathcal{E}_c} E_{c,j} - \sum_{j \in \mathcal{X}_c} X_{c,j}$$

$$I_{c,j} = \underbrace{\frac{Y^0_{c,j}}{Y^0_c}}_{\text{industry weight}} \underbrace{\frac{Y^1_{c,j} - Y^0_{c,j}}{Y^0_{c,j}}}_{\text{intensive growth}}, \quad E_{c,j} = \underbrace{\frac{Y^1_{c,j}}{Y^0_c}}_{\text{entry share}}, \quad X_{c,j} = \underbrace{\frac{Y^0_{c,j}}{Y^0_c}}_{\text{exit share}}$$

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The county-level coefficient inherits the same decomposition:

$$\beta^c = \underbrace{\frac{\sum_c z_c \sum_j I_{c,j}}{\sum_c z_c{}^2}}_{\beta^I} + \underbrace{\frac{\sum_c z_c \sum_j E_{c,j}}{\sum_c z_c{}^2}}_{\beta^E} - \underbrace{\frac{\sum_c z_c \sum_j X_{c,j}}{\sum_c z_c{}^2}}_{\beta^X}$$

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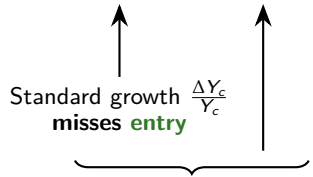
Condition for extensive margin irrelevance

$$\beta^c = \beta^I \text{ if and only if } \beta^E = \beta^X$$

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Channel 2: Most common growth rates miss full extensive margin

$$\Delta y_{ct} = \sum_j I_{c,j} + \sum_j E_{c,j} - \sum_j X_{c,j}$$



[Numerical example] [Back]

Channel 2: illustration

Changing the level changes the estimand mechanically, before any economics

		$t = 1$	$t = 2$	Growth rate
Firm level	Firm	50	80	

Channel 2: illustration

Changing the level changes the estimand mechanically, before any economics

		$t = 1$	$t = 2$	Growth rate
Firm level	Firm	50	80	

		$t = 1$	$t = 2$	Growth rate
Firm-market	m_1	25	30	
	m_2	0	50	
	m_3	25	0	
	Total	50	80	

Channel 2: illustration

Changing the level changes the estimand mechanically, before any economics

		$t=1$	$t=2$	Growth rate
Firm level	Firm	50	80	$\log(80/50) = 0.47$

		$t=1$	$t=2$	Growth rate
Firm-market	m_1	25	30	$\hat{g} = \log(30/25) = 0.18$
	m_2	0	50	undefined (entry)
	m_3	25	0	undefined (exit)
	Total	50	80	$\hat{g}_{\log} = 0.18$ ✗

(1) Log growth: only m_1 defined → drops 2 of 3 markets, misses the entrant that *is* the growth

Channel 2: illustration

Changing the level changes the estimand mechanically, before any economics

		$t=1$	$t=2$	Growth rate
Firm level	Firm	50	80	$+0.60$

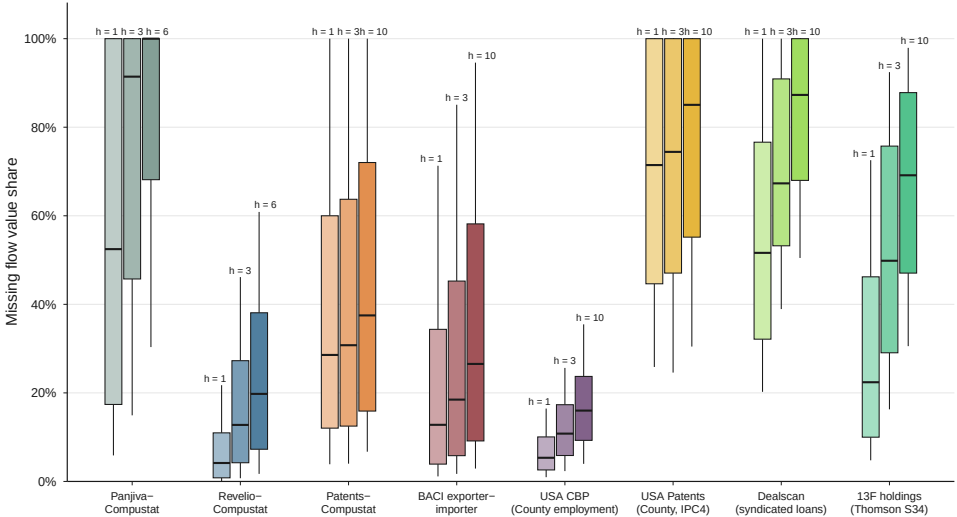
		$t=1$	$t=2$	Growth rate	Lag weight
Firm-market	m_1	25	30	$\frac{30-25}{25} = +0.20$	$w_1 = \frac{25}{50}$
	m_2	0	50	undefined (entry)	$w_2 = 0$
	m_3	25	0	$\frac{0-25}{25} = -1.00$	$w_3 = \frac{25}{50}$
	Total	50	80	$\hat{g}_{\text{std}} = -0.40$ ✗	

(2) Standard growth: equal *and* lagged weights both give -0.40; firm is $+0.60$
$\Rightarrow$ $+1.00$ gap because standard growth missed entrant

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Channel 2 in data: Extensive margin is large share of activity

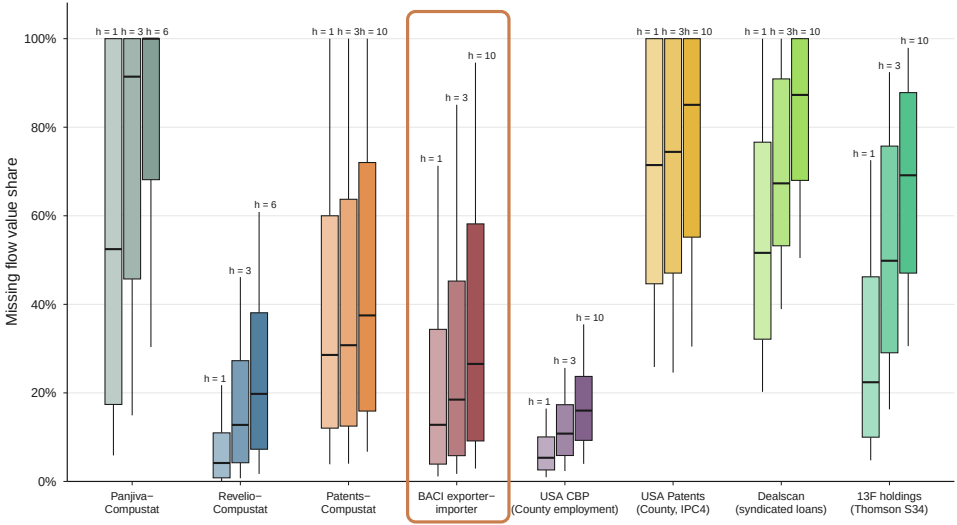
Share of activity in missing flows (zeros on log growth), by horizon h



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Channel 2 in data: Extensive margin is large share of activity

Share of activity in missing flows (zeros on log growth), by horizon h



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Channel 3: Jensen's inequality, numerical illustration

A treated firm's total exports are unchanged (100→100); only the composition shifts

Firm	Market	$t = 1$	$t = 2$	Log growth
Treated	US	20	60	$+1.099$
Treated	EU	80	40	-0.693
	Firm total	*100*	*100*	***0*** ✓
Control	US/EU	50/50	50/50	0

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	Firm total	*100*	*100*	***0*** ✓
Control	US/EU	50/50	50/50	0

EW: $\beta^{\text{f-m}}_{\text{OLS}} = +0.203$

WLS: $\beta^{\text{f-m}}_{w} = -0.334$

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	Firm total	*100*	*100*	***0*** ✓
Control	US/EU	50/50	50/50	0

EW: $\beta^{\text{f-m}}_{\text{OLS}} = +0.203$

WLS: $\beta^{\text{f-m}}_{w} = -0.334$

Different signs, none zero. Weighting does not fix Jensen: gap $\approx \frac{1}{2}\,\text{Var}_w(\Delta \log S_{imt})$

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Channel 3: Jensen's inequality in the wild

Suppose a city shock affects total employment

City-level economic development following access to banks (Fonseca and Matray, 2024, JFE)

In standard growth rates: $g_{\text{total}} = \omega_{\text{small}}\, g_{\text{small}} + \omega_{\text{large}}\, g_{\text{large}}$

In log growth: $\Delta \log \text{Emp}_{\text{total}} \neq \omega_{\text{small}}\, \Delta \log \text{Emp}_{\text{small}} + \omega_{\text{large}}\, \Delta \log \text{Emp}_{\text{large}}$

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Channel 3: Jensen's inequality in the wild

Suppose a city shock affects total employment

City-level economic development following access to banks (Fonseca and Matray, 2024, JFE)

Dep. var. (log)	# Firms	Emp. total	Emp. small	Emp. large	# Industries
	(1)	(2)	(3)	(4)	(5)
Treated × Post	0.098***	0.100***	0.214***	0.116***	0.047***
	(0.013)	(0.016)	(0.026)	(0.019)	(0.007)

No simple aggregation of columns (3) and (4) recovers column (2): the gap is mechanical

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Solution 1: MPGR in the entry/exit example

		$t=1$	$t=2$	Growth rate
Firm level	Firm	50	80	$+0.46$

		$t=1$	$t=2$	Growth rate
Firm-market	m_1	25	30	$+0.18$
	m_2	0	50	$+2.00$ (entry)
	m_3	25	0	-2.00 (exit)
	Total	50	80	

Midpoint growth rate: $\Delta^{mp} S_t = 2\dfrac{S_t - S_{t-1}}{S_t + S_{t-1}}$ $\rightarrow$ defined for entry and exit

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Solution 1: MPGR in the entry/exit example

		$t=1$	$t=2$	Growth rate
Firm level	Firm	50	80	$+0.46$ ✓

		$t=1$	$t=2$	Growth rate	Weight
Firm-market	m_1	25	30	$+0.18$	$w_1 = \frac{55}{130}$
	m_2	0	50	$+2.00$	$w_2 = \frac{50}{130}$
	m_3	25	0	-2.00	$w_3 = \frac{25}{130}$
	Total	50	80	$\Delta^{mp} S_i = +0.46$ ✓	

Midpoint growth rate: $\Delta^{mp} S_t = 2\dfrac{S_t - S_{t-1}}{S_t + S_{t-1}}$

Midpoint weight: $w_{im}^{mp} = \dfrac{S_{im,t} + S_{im,t-1}}{S_{i,t} + S_{i,t-1}}$ → weighted average matches firm-level

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MPGR in real data [Fonseca-Matray, 2024]

– Weight: $g^X_{j,c} / (\sum_{j \in c} g^X_{j,c}) \Rightarrow$ city level = city×industry level

$$\Delta Y_{c,j,t} = \beta_1 \ Treated_c \times Post_t + \delta_{j,p,t} + \varepsilon_{c,j,t}$$

Dependent Variable	Firms			Employment			Wage		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Treated×Post	0.104***	0.104***		0.094***	0.094***		0.025***	0.025***	
	(0.012)	(0.011)		(0.015)	(0.013)		(0.007)	(0.006)	
Match×Year FE	✓	✓		✓	✓		✓	✓	
Match×Industry×Year FE	–	–		–	–		–	–	
Observations	5,333	153,389		5,333	153,389		5,333	153,389	

Midpoint growth + proper weights ⇒ city-industry estimate aggregates exactly to the city level.

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MPGR in real data [Fonseca-Matray, 2024]

– Match×Industry×Year FE ⇒ industry shock ⊥ city-level reform

$$\Delta Y_{c,j,t} = \beta_1 \ Treated_c \times Post_t + \delta_{j,p,t} + \varepsilon_{c,j,t}$$

Dependent Variable	Firms			Employment			Wage		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Treated×Post	0.104***	0.104***	0.131***	0.094***	0.094***	0.102***	0.025***	0.025***	0.048***
	(0.012)	(0.011)	(0.013)	(0.015)	(0.013)	(0.014)	(0.007)	(0.006)	(0.007)
Match×Year FE	✓	✓	–	✓	✓	–	✓	✓	–
Match×Industry×Year FE	–	–	✓	–	–	✓	–	–	✓
Observations	5,333	153,389	153,389	5,333	153,389	155,038	5,333	153,389	153,389

Midpoint growth + proper weights ⇒ city-industry estimate aggregates exactly to the city level.

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MPGR in real data: trade [Matray, Mueller, Xu, Kabir, 2026]

Level: Origin×HS-4

Dependent variable	Δ Exports				
Level	O×HS-4				
	(1)	(2)	(3)	(4)	(5)
$EXIM_{p,o} \times Post_t$	-4.40				
	(1.57)				
	[0.0052]				
Fixed Effects					
Exporter×Year	✓				
Product (4-digit)×Year	✓				
Product (6-digit)×Year	–				
Product (6-digit)×Importer×Year	–				
Observations	109,199				

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MPGR in real data: trade [Matray, Mueller, Xu, Kabir, 2026]

Level: Origin×HS-6

- Define $A_{o,p,d,t} = (X_{o,p,d,t} + X_{o,p,d,t=pre}) \times 0.5$
- **Aggregation** with weights $A_{o,hs6,t} / (\sum_{hs6 \in [o,hs4,t]} A_{o,hs6,t})$

Dependent variable	Δ Exports				
Level	O×HS-4	O×HS-6			
	(1)	(2)	(3)	(4)	(5)
$EXIM_{p,o} \times Post_t$	-4.40	-4.40			
	(1.57)	(1.57)			
	[0.0052]	[0.0052]			
Fixed Effects					
Exporter×Year	✓	✓			
Product (4-digit)×Year	✓	✓			
Product (6-digit)×Year	–	–			
Product (6-digit)×Importer×Year	–	–			
Observations	109,199	8,419,512			

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MPGR in real data: trade [Matray, Mueller, Xu, Kabir, 2026]

Level: Origin×HS-6×Destination

- Define $A_{o,p,d,t} = (X_{o,p,d,t} + X_{o,p,d,t=pre}) \times 0.5$
- **Aggregation** with weights $A_{o,hs6,d,t} / (\sum_{hs6,d \in [o,hs4,t]} A_{o,hs6,d,t})$

Dependent variable	Δ Exports				
Level	O×HS-4	O×HS-6	O×HS-6×Dest		
	(1)	(2)	(3)	(4)	(5)
EXIM$_{p,o}$×Post$_t$	-4.40	-4.40	-4.40		
	(1.57)	(1.57)	(1.57)		
	[0.0052]	[0.0052]	[0.0052]		
Fixed Effects					
Exporter×Year	✓	✓	✓		
Product (4-digit)×Year	✓	✓	✓		
Product (6-digit)×Year	–	–	–		
Product (6-digit)×Importer×Year	–	–	–		
Observations	109,199	8,419,512	23,775,713		

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MPGR in real data: trade [Matray, Mueller, Xu, Kabir, 2026]

Level: Origin×HS-6×Destination

- Define $A_{o,p,d,t} = (X_{o,p,d,t} + X_{o,p,d,t=pre}) \times 0.5$
- **Additional fixed effects**: cleanly compare unobserved heterogeneity at more aggregate levels

Dependent variable	Δ Exports				
Level	O×HS-4	O×HS-6	O×HS-6×Dest	O×HS-6×Dest	O×HS-6×Dest
	(1)	(2)	(3)	(4)	(5)
$EXIM_{p,o} \times Post_t$	-4.40	-4.40	-4.40	-4.16	-5.02
	(1.57)	(1.57)	(1.57)	(1.64)	(2.40)
	[0.0052]	[0.0052]	[0.0052]	[0.011]	[0.037]
Fixed Effects					
Exporter×Year	✓	✓	✓	✓	✓
Product (4-digit)×Year	✓	✓	✓	–	–
Product (6-digit)×Year	–	–	–	✓	–
Product (6-digit)×Importer×Year	–	–	–	–	✓
Observations	109,199	8,419,512	23,775,713	23,775,713	23,775,713

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Solution 2: Poisson regression

Estimate sales in *levels* with firm (or firm-market) fixed effects:

$$E(S_{imt} \mid i, m, t) = \exp(\alpha_{im} + \mu_t + \beta_d \, Treated_i \times Post_t)$$

- Disaggregation invariance by construction: the first-order condition aggregates naturally over markets
- Zeros are not a problem (Poisson handles them natively)
- Does not deliver aggregation invariance (Part II)

Key takeaway

Midpoint growth and Poisson are the two consistent estimators for disaggregation

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A flexible multi-level data-generating process

- **Simulated data:** we need a DGP that accommodates
  - multiple nested aggregation levels
  - entry and exit (i.e., zeros)

- We build on Eaton, Kortum, and Kramarz (2011): exporters selling different products across destinations
  - Structure is general: any unit whose activity aggregates over sub-units, treatment shifting activity *within* units and the *set* of active units
    - Banks allocating credit across borrowers; multi-establishment firms across locations; firms expanding employment across occupations
    - Multi-product scope; global sourcing; multinational affiliates; supplier–buyer networks

- **Two-level, two-period design**
  - firms i sell in markets m
  - $t = 1$ (pre) and $t = 2$ (post); half the firms treated

Treatment is a productivity shock on both margins

Treatment = proportional productivity increase, randomly assigned:

$$\theta_{i,2}^{Treated} = \theta_{i,1}(1+g), \qquad \theta_{i,2}^{Control} = \theta_{i,1}$$

Total firm sales aggregate across all markets:

$$E(S_{it} \mid \theta_{it}) = \underbrace{\left(1+\lambda_0\,\theta_{it}^{\eta}\right)}_{\text{Number of markets}} \underbrace{\mu_0\,\theta_{it}^{\gamma}}_{\text{Sales per market}}$$

Extensive margin

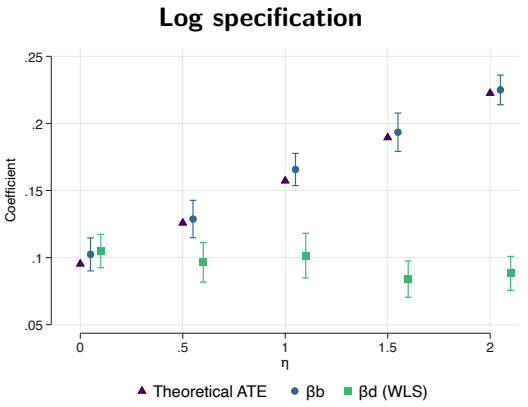
- More productive firms operate in more markets
- Parameter η governs how strongly market participation responds to productivity

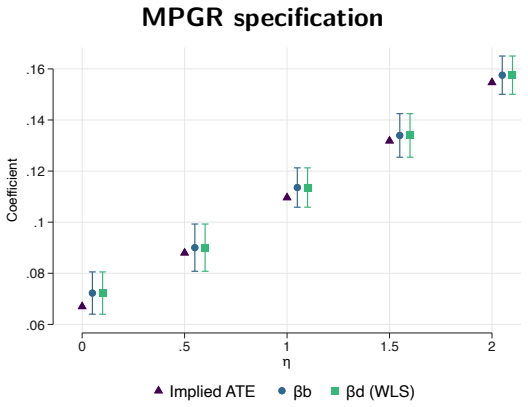
Intensive margin

- More productive firms sell more per market
- Parameter γ governs how strongly productivity translates into market-level revenue

Simulation: entry and exit

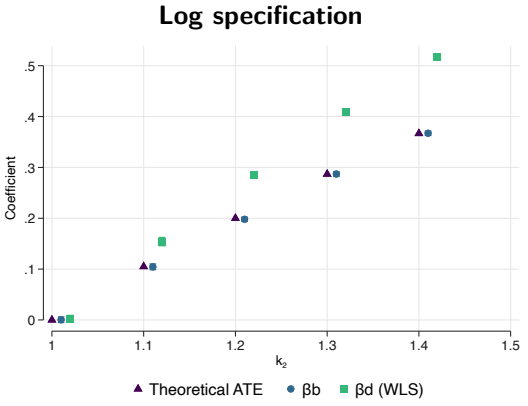
Treatment induces greater expansion into new markets as η (x-axis) increases. Using weights and homogeneous effects shuts down implicit weights and Jensen's inequality

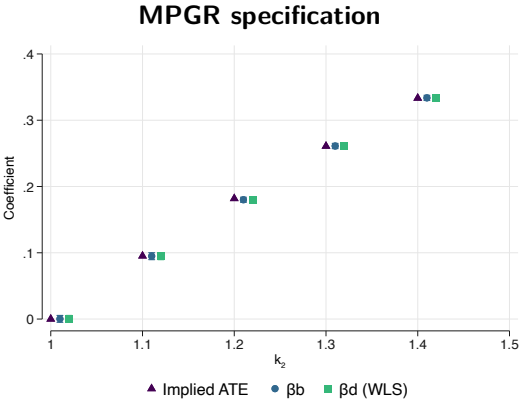




Simulation: Jensen's inequality

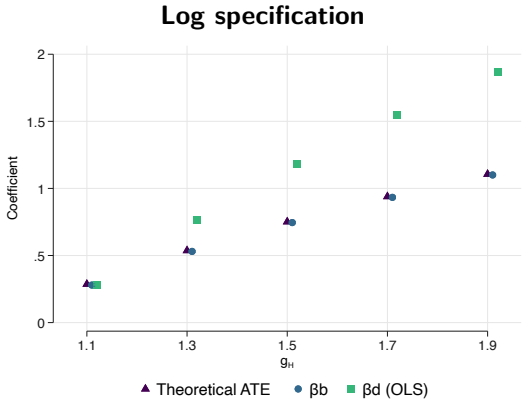
All firms operate in the same number of markets, so there is no extensive margin and no implicit-weights channel. Firm-market sales are Gamma draws, $S_{imt} \sim \Gamma(k_t, \lambda_t)$. Treatment increases k_2 (x-axis), which changes treated within-firm dispersion of log growth

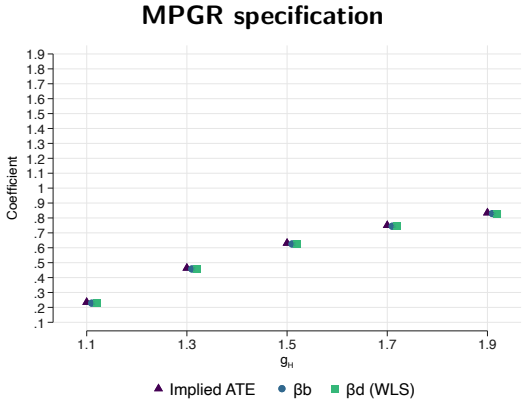




Simulation: implicit weights

Ex ante more productive firms operate in more markets, removing the extensive-margin channel. Treatment is a productivity upgrade that depends on initial productivity, with $g_H > g_L$, so treatment covaries with the number of disaggregated units.





Practical guidance: Disaggregation

1. **Is there entry/exit at the sub-unit level?**
   → Use midpoint growth rate or Poisson

2. **Are treatment effects heterogeneous across firms?**
   → Weight appropriately (midpoint weights or firm-market FE)

3. **Using log growth?**
   → Compare with midpoint specification as robustness

Key takeaway

When in doubt, use the midpoint specification: it aggregates linearly within cell, so it gives the same answer when the gap is small and the *disaggregation-invariant* answer when it isn't

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Midpoint weights vs. lagged weights

- Midpoint weights: $w_{im}^{mp} = \frac{S_{im,t}+S_{im,t-1}}{S_{i,t}+S_{i,t-1}}$ — depend on post-treatment outcomes
- Lagged weights: $w_{im}^{s} = \frac{S_{im,t-1}}{S_{i,t-1}}$ — predetermined
- Under firm-level random assignment, midpoint weights are innocuous: the disaggregated regression targets the same estimand as the baseline
- **Diagnostic:** check $\hat{\beta}$ is unchanged under lagged weights w^s (the direct test). A high w^{mp}-w^s correlation is suggestive but does not by itself bound the bias
- **Fallback:** standard growth with lagged weights (intensive margin) + separate entry regression (extensive margin)

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Alternative example: two firms, two banks

	Firm A (treated)		**Firm B** (control)	
	Bank 1	Bank 2	Bank 1	Bank 2
Credit $t-1$	100	0	80	20
Credit t	80	30	75	25
Log growth	-0.22	undefined	-0.06	$+0.22$
MPGR	-0.22	$+2.00$	-0.06	$+0.22$

- Firm A opens a new bank relationship (Bank 2): log growth drops it
- Firm-level credit growth: Firm A $= +10\%$, Firm B $= 0\% \Rightarrow \hat{\beta} = +10\%$
- Firm-bank log growth (dropping entry): $\hat{\beta}_{\text{disagg}} = \frac{-0.22-(-0.06+0.22)/2}{1} = -0.30$
- Firm-bank MPGR with midpoint weights: $\hat{\beta}_{\text{disagg}} = +10\%$ ✓

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Extensive margin decomposition

Midpoint growth decomposes into intensive and extensive margins:

$$2\frac{S_{it} - S_{i,t-1}}{S_{it} + S_{i,t-1}} = \underbrace{2\frac{\text{Entry}_i - \text{Exit}_i}{S_{it} + S_{i,t-1}}}_{\text{extensive margin}} + \underbrace{2\frac{\text{Int}_t - \text{Int}_{t-1}}{S_{it} + S_{i,t-1}}}_{\text{intensive margin}}$$

- Disaggregation invariance holds if treatment only affects the intensive margin
- Test: regress extensive-margin component on $Treated_i$
- Large coefficient $\rightarrow$ extensive margin matters; disaggregated specs miss part of the effect

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Why not just cluster standard errors?

- This paper is about the **estimand**, not about standard errors
- Clustering fixes inference (Moulton, 1990), but the point here is that the coefficient itself targets a different object at different aggregation levels
- You can have perfectly valid standard errors around a biased estimate of what you think you're measuring

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Why does $\delta^{sp} = 0$ test aggregation invariance?

Assumption (aligned specifications). Use the same predetermined weights and controls at both levels, with

$$y_j \equiv \sum_{i \in j} \omega_{ij} y_i, \qquad \textit{Share Treated}_j \equiv \sum_{i \in j} \omega_{ij} \textit{Treated}_i, \qquad \sum_{i \in j} \omega_{ij} = 1$$

Start from the augmented firm-level regression:

$$y_i = \alpha + \beta^b \, \textit{Treated}_i + \delta^{sp} \, \textit{Share Treated}_j + \epsilon_i$$

Aggregate within industry j:

$$\begin{aligned} y_j &= \sum_{i \in j} \omega_{ij} y_i \\ &= \alpha + \beta^b \sum_{i \in j} \omega_{ij} \textit{Treated}_i + \delta^{sp} \sum_{i \in j} \omega_{ij} \textit{Share Treated}_j + \sum_{i \in j} \omega_{ij} \epsilon_i \\ &= \alpha + (\beta^b + \delta^{sp}) \, \textit{Share Treated}_j + \epsilon_j \end{aligned}$$

$$\Rightarrow \quad \hat{\beta}^{\text{industry}} = \hat{\beta}^b + \hat{\delta}^{sp} \quad \Rightarrow \quad \hat{\delta}^{sp} = 0 \;\Rightarrow\; \hat{\beta}^b = \hat{\beta}^{\text{industry}}$$

β^b is the augmented (within-industry) slope. The raw firm coefficient β^{firm} equals it only at the null $\delta^{sp} = 0$ (Frisch-Waugh: off the null β^{firm} mixes the within and between slopes), so $\delta^{sp} = 0$ is a test at the null. This is an algebraic mapping only: a nonzero δ^{sp} need not be causal spillovers if $\textit{Share Treated}_j$ is correlated with industry-level confounders or heterogeneous direct effects

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Relation to the zeros literature

- (Cohn, Liu, and Wardlaw, 2022) and (Chen and Roth, 2024): common transformations ($\log(1+x)$, arcsinh) yield biased or hard-to-interpret estimates

- Complementary, not competing:
  - The zeros literature: what transformation to use when Y has zeros
  - This paper: what happens when you *change the level of aggregation*

- Zeros are one of three channels (entry/exit), but weights and Jensen are distinct issues that arise even without zeros

- Midpoint growth addresses zeros *and* the other two channels simultaneously

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Heterogeneous treatment effects and sorting

- If treatment effects vary across firms, and firms sort into industries based on characteristics correlated with responsiveness, aggregation invariance can fail even under aligned specifications

- **Example:** financially constrained firms respond less to treatment; industries with more treated firms also have more constrained firms
  $\rightarrow$ Industry-level regression spuriously suggests a weak aggregate effect

- **Fix:** augment the firm-level regression with interaction terms:

$$\Delta \log S_i = \alpha + \beta^b \, Treated_i + \delta^{sp} \, Share\ Treated_j + \delta^{het} \, Treated_i \times X_i + \epsilon_i$$

  where X_i captures the heterogeneity dimension (e.g., financial constraints)

- Can also detect heterogeneous spillovers (Berg, 2021): does the spillover effect differ by firm characteristics?

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Backup

Numerical example

Two states, four equal counties each. No entry, no Jensen, no weights.

$$Y_c = 4\,Treated_c + 4\,Share\ Treated_{s(c)} \qquad \text{direct effect} = 4, \quad \text{spillover} = 4$$

	Treated counties	$Share\ Treated_s$	Y treated	Y control
State A	1	1/4	5	1
State B	3	3/4	7	3

Raw comparison: $\beta^{state} = 8$ and $\beta^{county} = 5$, so $\beta^{state} - \beta^{county} = 3 \neq$ true spillover 4

Alignment problem: the county regression omits $Share\ Treated_s$. Treated counties are more common in State B, so β^{county} already contains $\pi \times 4 = 1$ point of the spillover

Fix: add state exposure to the county regression

$$Y_c = \alpha + \beta^{county,aug}\ Treated_c + \delta^{spill}\ Share\ Treated_{s(c)} + \varepsilon_c$$

$$\beta^{county,aug} = 4, \qquad \delta^{spill} = 4$$

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Backup: pinning down the weights

- NB: exposure is a *weighted share*, $\textit{Share Treated}_j = \sum_{i:\, j(i)=j} \omega_{ij}^s \, \textit{Treated}_i$, weights $\omega_{ij}^s = S_{i,t-h} / S_{j,t-h}$ summing to one

- Two requirements pin down the weights ω_{ij} in $\textit{Share Treated}_j$:
  1. match the weights that aggregate *growth* on the left-hand side
  2. sum to one within each industry (industries counted equally)

- For the standard (and midpoint) growth rate:

$$\omega_{ij}^s = \frac{S_{i,t-h}}{S_{j,t-h}}, \qquad \omega_{ij}^{mp} = \frac{S_{it} + S_{i,t-h}}{S_{jt} + S_{j,t-h}}$$

- Check: $\underbrace{\sum_i \omega_{ij}^s}_{=1}$

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A simulation to size the mechanical channels

- Firm productivity drawn Pareto (tail α governs how concentrated the industry is)
- Industry treatment shares *Share Treated*$_j$ vary across industries
- Entry probability $p_{\text{entry},j} = \kappa \cdot \textit{Share Treated}_j$ with $\kappa = 0.375$; about 50 potential entrants per industry
- Incumbent spillover multiplier $\phi_j = 1 - \psi \cdot \textit{Share Treated}_j$ with $\psi = 0.25$

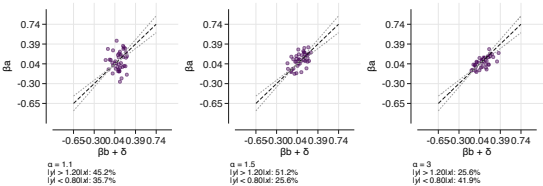
Parameters typed to match the paper text ($\kappa = 0.375$).

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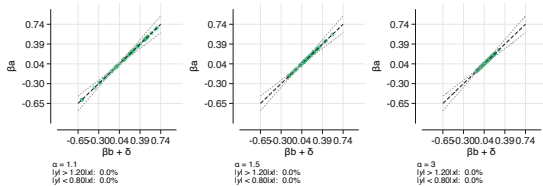
Implicit weights matter when industries are concentrated

– Plots the augmented firm coefficient (x) against the industry coefficient (y); dashed lines at $\pm 10\%$

(a) Log growth rate – OLS



(b) Standard growth rate – WLS



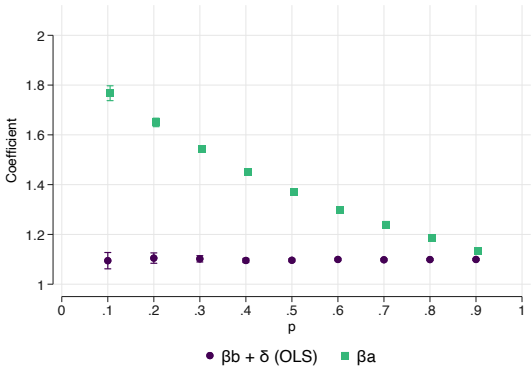
In the figures $\beta^a = \beta^{state}$, $\beta^b = \beta^{county}$.

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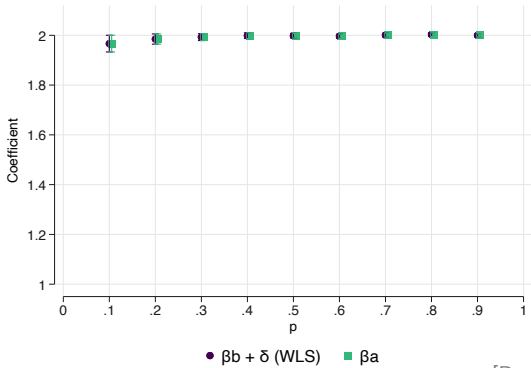
Jensen matters at intermediate treatment shares

– Concavity term $C_{jt} \approx \frac{1}{2}\left(\beta^2 \, \textit{Share Treated}_j(1 - \textit{Share Treated}_j) + \mathrm{Var}_\omega\right)$: largest at intermediate shares, vanishes as the share $\to 1$

(a) Log growth rate – OLS



(b) Standard growth rate – WLS



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In the figures $\beta^a = \beta^{state}$, $\beta^b = \beta^{county}$.

Detecting spillovers: log/OLS can mask offsetting margins

– One calibration where entry *offsets* incumbent spillovers: an existence proof, not a generic result

→ Panel A (log / OLS): $\hat{\beta}^a \approx \hat{\beta}^b$ can read as a false "no spillovers"; Panel B (standard / WLS): a large negative $\hat{\delta}$ offset by a positive $\hat{\delta}^E$

Specification	Mean	SE
Panel A: Log growth rate – OLS		
$\hat{\beta}^a$	0.0268	0.0088
$\hat{\beta}^b$	0.0299	0.0051
Panel B: Standard growth rate – WLS		
$\hat{\beta}^a$	0.0265	0.0090
$\hat{\beta}^b + \hat{\delta} + \hat{\beta}^E$	0.0265	0.0090
$\hat{\delta}$	-0.2405	0.0089
$\hat{\beta}^b$	0.0867	0.0041
$\hat{\beta}^E$	0.1803	0.0027

$\beta^a = \beta^{state}$, $\beta^b = \beta^{county}$. $\hat{\delta} = -0.24$ hides in a near-zero industry coefficient; $\hat{\beta}^b$ triples ($0.03 \to 0.09$), so Panel A matches two log-contaminated numbers.

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Midpoint fails

- Midpoint growth restored disaggregation invariance. For aggregation it breaks

- Its weights depend on *contemporaneous* sales $S_{it} + S_{i,t-h}$, which treatment moves

$\rightarrow$ *Share Treated*$_j$ becomes a function of the outcome $\Rightarrow$ simultaneity bias

  $\approx$ "Bad-control bias:" midpoint weights are themselves affected by treatment
  (Angrist and Pischke, 2009)

- No such issue when disaggregating: treatment is **fixed** at the firm level

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Poisson fails

- Poisson also restored *disaggregation* invariance; at the industry level it breaks too

- Firm-level Poisson:

$$E(S_{it} \mid i, t) = \exp(\alpha_i + \mu_t + \beta_b \, Treated_i \times Post_t)$$

- Two-firm industry (0 untreated, 1 treated):

$$E(S_{jt} \mid j, t) = \exp(\mu_t) \left[\exp(\alpha_0) + \exp(\alpha_1 + \beta_b \, Post_t)\right]$$

$\rightarrow$ the treatment component is governed by $\dfrac{\exp(\alpha_1)}{\exp(\alpha_0) + \exp(\alpha_1)}$

  - depends on the fixed effects, not on *Share Treated*$_j$ alone

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The estimand problem

$\beta^{state} - \beta^{county} = \delta^{spill} + \delta^{est} + \delta^{wedge} + \delta^{E}$

- Firm: $\Delta S_{it} = \alpha + \beta^{county}\, Treated_i + e_{it}$; Industry: $\Delta S_{jt} = \alpha + \beta^{state}\, Share\ Treated_j + e_{jt}$

→ Regressor changes, industry regression needs its *own* exogeneity, not inherited:

$$E[e_{it}\, Treated_i] = 0 \not\Rightarrow E[e_{jt}\, Share\ Treated_j] = 0$$

- δ^{est} has two sources:

$$\delta^{est} = \underbrace{\delta^{endo}}_{\text{endogenous assignment}} + \underbrace{\delta^{comp}}_{\text{composition}}$$

  - δ^{endo}: treatment responds to industry shocks (e.g. targeting distressed industries)
    → assume away (else "spillover" undefined)
  - δ^{comp}: heterogeneity × exposure, present *even with* $\delta^{endo} = 0$ [FWL derivation] [Controlling]
    - True under homogeneous treatment or additional controls such that:

$$\mathrm{Cov}_{\omega}(\tau_i,\ Share\ Treated_j \mid Treated_i = 1) = 0$$

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Appendix: the estimand problem, derived

- Assume $\delta^{endo} = 0$ (industry assignment exogenous); FWL then isolates the composition part δ^{comp}
- Put *both* regressors in one firm-level regression (defines δ):

$$\Delta S_{it} = \alpha + \beta^{county}\, Treated_i + \delta\, Share\ Treated_j + \varepsilon_{it}$$

- Frisch–Waugh–Lovell: partition on $\{1,\ Treated_i\}$, so δ loads on the residualized share

$$\widetilde{ST}_j = Share\ Treated_j - \mathrm{E}_\omega[Share\ Treated_j \mid Treated_i]$$

- The linear projection splits it in two:

$$\delta = \underbrace{\delta^{spill}}_{\text{causal response to others}} + \underbrace{\delta^{comp}}_{\text{heterogeneity} \times \text{exposure}}$$

→ *Maintained condition* sets $\delta^{comp} = 0$ (τ_i = firm i's effect; across treated firms, regression weights; holds under homogeneous effects):

$$\mathrm{Cov}_\omega(\tau_i,\ Share\ Treated_j \mid Treated_i = 1) = 0 \ \Rightarrow\ \delta = \delta^{spill}$$

→ Limitation: δ is a *linear* projection, so a spillover with zero linear projection is not detected [Back]

Backup: inference is preserved across levels

– Under industry-level clustering, the standard error of $\beta^{county} + \delta$ equals the standard error of β^{state}

$\rightarrow$ comparing coefficients across levels and estimating δ directly deliver the *same* inference

Heterogeneous treatment complicates detection

- Suppose constrained firms react less, and $Corr(Treated, Constrained) > 0$
  - industries with high $Share\ Treated_j$ also have a high share constrained
  - then δ^{comp} can be < 0 *without any spillover*: $\delta = \delta^{spill} + \delta^{comp}$

- To recover δ^{spill}, control for the source of heterogeneity:
  - $Constrained$, $Treated \times Constrained$, $Share\ Constrained$, $Avg\ Treated \times Constrained$

$\rightarrow$ but the researcher must know the source of heterogeneity

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